

THE TRIALS OF XANTHIUS

Among the ancient Greeks, the people of Athens led all others in their mental acuity. The gods created a series of trials to test the Athenians' reasoning ability, as well as their courage (and perhaps to amuse themselves). As an incentive, they provided a fabulous treasure to be won by whoever was successful in passing every trial.

The trials involved following a path through a forest with choices to be made at three points. Once the challenge was accepted, there was no turning back and no retracing of steps at any time. Dire consequences awaited a challenger who made an incorrect judgment.

No citizen of Athens desired to accept the risk until Xanthius, descendant of Bellerophon (the slayer of the Chimera), accepted the challenge.

1 The First Trial

Hardly had Xanthius entered the forest on the designated path when it branched into two. He was told that this was the first trial and that a sign at each path gave instructions. However, Xanthius was informed that at least one of the signs was false. The signs read as follows:

- A. THIS PATH DOES NOT LEAD TO THE SECOND TRIAL.**
- B. THE SIGN AT PATH A IS TRUE.**

Should Xanthius travel on Path A, Path B, do both lead him astray, or do both lead him to the second trial?

2 The Second Trial

Xanthius chose correctly and proceeded farther into the forest before he came to a branching of the path again, but this time into three paths. He was informed that of the signs at the three paths, exactly one was false, but that at least one of the paths leads to the next trial. The signs read as follows:

- A. THIS SIGN IS THE FALSE SIGN OR PATH C'S SIGN IS THE FALSE SIGN.**
- B. PATH A DOES NOT LEAD TO THE THIRD TRIAL.**
- C. THIS PATH LEADS TO THE THIRD TRIAL.**

Which of the paths can Xanthius choose safely, and which (if any) of the paths must Xanthius avoid?

3 The Third Trial

Again, Xanthius chose correctly and, after proceeding into the forest for some time, the path once more branched into three paths. He was informed that exactly one of the signs at the three paths was false, but he was also told that an extra danger had been kept from him: at most one of the three paths would lead to the treasure, but possibly none do! The signs read as follows:

A. PATH B LEADS TO THE TREASURE.

B. IF PATH A DOES NOT LEAD TO THE TREASURE, THEN PATH C DOES.

C. PATH A DOES NOT LEAD TO THE TREASURE, AND PATH B DOES NOT LEAD TO THE TREASURE.

Is there a path that leads Xanthius to the treasure? If so, which one? If not, why not?